A Soft Sensor for Multirate Quality Variables Based on MC-CNN

Bing Song, Yichen Zhou, Hongbo Shi, Yang Tao, and Shuai Tan

***Abstract*— In recent years, data-driven soft sensor modeling methods have been widely used in industrial production, chemistry, and biochemical. In industrial processes, the sampling rates of quality variables are always lower than those of process variables. Meanwhile, the sampling rates among quality variables are also different. However, few multi-input multi-output (MIMO) sensors take this temporal factor into consideration. To solve this problem, a deep-learning (DL) model based on a multitemporal channels convolutional neural network (MC-CNN) is proposed. In the MC-CNN, the network consists of two parts: the shared network used to extract the temporal feature and the parallel prediction network used to predict each quality variable. The modified BP algorithm makes the blank values generated at unsampled moments not participate in the backpropagation (BP) process during training. By predicting multiple quality variables of two industrial cases, the effectiveness of the proposed method is verified.**

***Index Terms*— Convolutional neural network (CNN), deep learning (DL), multirate sampling, quality variables, soft sensor.**

I. Introduction

IN THE industrial processes, the key quality variables play an important role in efficiency, safety, energy saving, and environmental protection [1], [2], [3]. However, most quality variables are usually analyzed offline due to the complex industrial environment, expensive analytical instruments, or high-latency and low-reliability data with low sampling rate rather than measure these quality variables online [4], [5], [6]. On the contrary, the process variables are measured at low cost in the condition of high sampling rate and low requirements, and these process variables often have a strong relationship with the quality variables. Thus, mathematical models can be built through process variables to predict quality variables which are hard to measure online. Based on this, the concept of soft sensor has been proposed and has been successfully utilized in industrial production, chemical, biochemical, and other processes [7], [8], [9]. According to the modeling principle, soft sensors can be divided into two different categories: first-principles models (FPMs) [10] and data-driven models (DDMs) [11].

FPMs are constructed based on the mechanistic relationship between quality variables and process variables, which require sufficient theoretical basis and empirical support [12]. However, as the complexity of industrial processes increases, it becomes even more difficult to acquire industrial knowledge. At the same time, due to the development of technology making data collection and storage more convenient, data-driven soft sensors have received more and more attention [13], and the advantages of data-driven soft sensors such as high efficiency, low cost, and low environmental requirements have caused data-driven soft sensors to be widely applied in practical production.

Some classical data-driven soft-sensor models, such as least absolute shrinkage and selection operator (LASSO) [14], partial least-squares (PLSs) regression [15], support vector regression (SVR) [16], cannot cope with a large number of process variables with strong nonlinearity. In contrast, neural networks can be activated by functions like ReLU and show strong power in handling large-scale, high-dimensional, and nonlinear regression problems [17]. In recent years, deep learning (DL) has been developed, such as deep belief networks (DBNs) [18], variational autoencoders (VAEs) [19], recurrent neural networks (RNNs) [20], autoencoders (AEsciteref21, and convolutional neural networks (CNNs) [22]. The DL methods have the superior capability to extract features from the high-dimensional data of industrial processes with strong correlations and high redundancy, which means that the DL methods have more potential for soft sensor modeling. Among them, the CNN can extract features on both spatial and temporal scales and construct the regression models by connecting the relationship between the feature and quality variables. Bargellesi et al. [23] proposed a DL-based approach by utilizing a special architecture containing AEs and extended convolution to construct soft sensing models and show its effectiveness in a real case study. Wang et al. [24] integrated model trained by sufficient process data and that trained by limited pulse response for a dynamic process.

There are many effective ways to model soft sensors, but most of them put emphasis on the prediction of only one quality variable [25], called multi-input–single-output (MISO) soft sensors, and there is little research on

Manuscript received 22 March 2023; revised 2 August 2023 and 30 October 2023; accepted 28 November 2023. Date of publication 23 February 2024; date of current version 6 August 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62073140, Grant 62073141, and Grant 62103149; in part by the Shanghai Rising-Star Program under Grant 21QA1401800, in part by the Shanghai Chenguang Project under Grant 21CGA37; and in part by the National Key Research and Development Program of China under Grant 2020YFC1522502 and Grant 2020YFC1522505. *(Corresponding author: Bing Song.)*

The authors are with the Key Laboratory of Smart Manufacturing in Energy Chemical Process of the Ministry of Education, East China University of Science and Technology, Shanghai 200237, China (e-mail: songbing@ecust.edu.cn).

Digital Object Identifier 10.1109/TNNLS.2024.3360030

2162-237X © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

multi-input–multi-output (MIMO) soft sensors. Even more, some MIMO models are only constructed to predict multiple quality variables with the same sampling rate [26]. Zhou et al. [27], [28] proposed several methods for multirate process data to effectively utilize the collected data in fault detection. Shen et al. [29] turned the process variables at multiple sampling rates into resolution maps through VAE and then used a feature pyramid to predict multiple quality variables with the same sampling rate to build MIMO soft sensors.

However, few MIMO soft sensors can predict quality variables with different sampling rates at the same time. To solve the problem, Yuan et al. [30] proposed the MR-SAE model. MR-SAE first extracted features of process variables through SAE, then built a shared network for fine-tuning, and finally conducted multiple networks to train every quality variable and predicted multiple quality variables at multiple sampling rates. Moreover, most industrial processes have temporal characteristics due to the feedback control and physicochemical reaction characteristics. There are strong correlations between process variables and the temporal dependency of quality variables on process variables. Ignoring these correlations in soft sensor modeling will greatly influence the effect of models. For some industrial processes, ignoring these correlations may even make the opposite trend for quality variables prediction, which has a great influence on process monitoring. Hence, it is important to take the temporal correlations between quality variables and process variables into consideration when soft sensor modeling [31], [32]. However, for quality variables with low multiple sampling rates, it is hard for the traditional MIMO model to introduce the temporal correlations between variables into the model. At the same time, all quality variables cannot be guaranteed to be sampled at the same moment, but eliminating quality variables data not being sampled will cause serious data wasting. Moreover, in the case of a small amount of data, it is hard to use the limited real values to train the model and then predict quality variables online. Therefore, an MIMO soft sensor that can use all the available data and learn the temporal correlations between quality variables and process variables with different sampling rates is needed.

Based on the problems above, a DL model named multitemporal channel CNN (MC-CNN) is proposed for soft sensing of multirate quality variables. In the MC-CNN, the network is divided into the shared networks and the parallel prediction network. In the shared network, the combination of fully connected layers and convolutional layers extracts features by using multitemporal channels. The network can learn the temporal correlations between quality variables and process variables through moving time windows even if the sampling rates of quality variables are apparently lower than those of process variables. The parallel prediction network is composed of each quality variable prediction network, and the feature extracted from the shared network is used as the input to each subnetwork. Finally, the whole prediction model is constructed to predict all quality variables at a high frequency through corresponding subnetworks. Therefore, a more effective MIMO soft sensor can be constructed to predict multirate quality variables. In addition, the difference between MR-SAE and MC-CNN can be reflected in two aspects. First, MR-SAE emphasizes learning the mutual relationship among process variables and uses SAE to extract features at first. In contrast, MC-CNN emphasizes learning the temporal correlations between quality variables and process variables, and SAE is not used in MC-CNN. Second, in the shared network, MC-CNN uses the structure of multitemporal channels to extract the temporal feature, and MR-SAE only extracts features through a single channel.

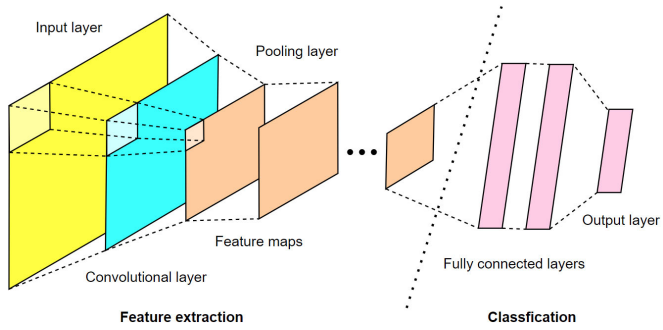

Fig. 1. Structure of the CNN.

The contributions of the article are listed as follows.

1) A novel soft sensor modeling method MC-CNN is proposed to predict multiple quality variables with different sampling rates.

2) MC-CNN proposed in the article utilizes the structure of multitemporal channels. The structure can learn the temporal correlations between quality variables and process variables, which makes it the model to track the temporal prediction curves and achieve satisfactory performance.

3) MC-CNN can make full use of low-sampling quality variables and high-sampling process variables through multitemporal channels and predict quality variables at a high frequency.

The rest parts of this article are organized as follows. First, the preliminaries are briefly introduced in Section II. Then, the proposed MC-CNN method is described in detail in Section III. Moreover, Section IV verifies the effectiveness of the proposed method under an industrial process platform and the actual energy storage device. Finally, conclusions are summarized in Section V.

II. Preliminaries

A. Convolutional Neural Network

CNN consists of two parts. As shown in Fig. 1, the left part is responsible for feature extraction, composed of an input layer, a convolutional layer, and a pooling layer, and the right one is generally composed of multiple fully connected layers. In this part, the feature obtained in the pooling layer serves as input to the fully connected layer and this part performs the classification task [33]. The model can be trained by comparing the predicted result with the actual label.

B. BP Algorithm

The backpropagation (BP) algorithm consists of two processes: forward propagation and BP of the network. The input

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

starts from the input layer to the output layer to compare the predicted values with the real values during the forward propagation. During the BP, the error between the predicted output and the real values is returned by the path of the network, dispersing the error to the units in each layer [34]. Through the chain rule, the error partial derivative of the units in each layer can be solved layer by layer and then the parameters are adjusted to make the final prediction output closer to the real values. Therefore, the loss function of the network can be defined as

$$J\left(W, b; x_{(i)}, y_{(i)}\right) = \frac{1}{2N}\sum_{i=1}^{n}\left\{f\left(x_{(i)}\right) - y_{(i)}\right\}^2 \tag{1}$$

where N denotes the batch size, W and b denote the weights and bias of the network, respectively, $x_{(i)}$ denotes the input of the network, $y_{(i)}$ denotes the label, and $f(x_{(i)})$ denotes the predicted values from the output of network. The iteration of the network can be completed by reducing the total loss of (1). More details can be found in [35].

III. Multitemporal Channel CNN

In this section, an MC-CNN is proposed. It solves the problem of predicting multirate quality variables by introducing the temporal feature of process variables extracted by the multitemporal channel into the model and increasing the utilization of real-value data. This section contains three parts: 1) data form and data preprocessing; 2) structure and training procedure of the MC-CNN; and 3) procedure of the MC-CNN for quality prediction.

A. Data Form and Data Preprocessing

In this article, multiple quality variables can have different sampling rates. In addition, process variables have a significantly higher sampling rate than quality variables. Assume that the sampling rates of Y_1 and Y_2 are 6 and 8, respectively. As can be seen from (2), within 24 moments of a sampling cycle, Y_1 and Y_2 only has real values at 4 and 3 moments, respectively. Therefore, if traditional MIMO models are used to train, such as SAE, it will have the problem that the real values of quality variables for training are insufficient. What is more, traditional MIMO models cannot extract the temporal feature of variables

$$\begin{aligned} Y_1 &= \{y_1(1), y_1(7), y_1(13), y_1(19), y_1(25)\}, \\ Y_2 &= \{y_2(1), y_2(9), y_2(17), y_2(25)\}. \end{aligned} \tag{2}$$

Moreover, to enable the input process variable and the output quality variable to meet the requirements of the designed network, the dataset needs to be preprocessed. In the article, the first step is to standardize the data through the Z-score standardization method. The second step is to deal with blank values so that the blank values will be not involved in the BP process of the network. The third step is to generate an independent dataset with dimension-enhanced data after saving the standardized dataset. The way to enhance the dimension of data is by splicing the time window. The dimension of the new dataset depends on the number of multitemporal channels M. M changes with different industrial processes. In this article, M is set to 5. Assume the dataset completing normalization and blank values processing can be expressed as

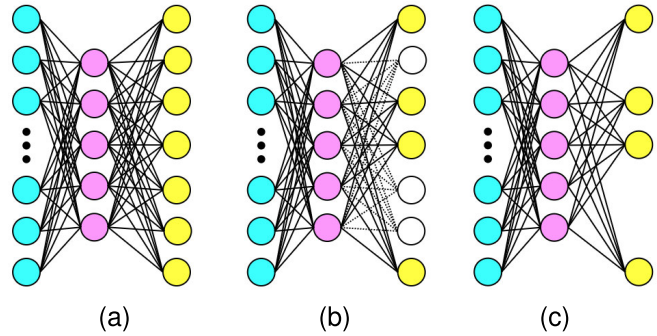

Fig. 2. (a) Common neural network. (b) Neural network with blank units and corresponding channels. (c) Neural network excluding blank units and corresponding channels.

$$\begin{aligned} X_{nb} &= [x_1, x_2, \ldots, x_T]^T \in R^{T\times m}, \\ Y_{nb} &= \left[y_1, y_2, \ldots, y_T\right]^T \in R^{T\times n} \end{aligned} \tag{3}$$

after enhancing the dimension of the data, the dataset can be expressed as

$$\begin{aligned} D_x &= \begin{bmatrix} \left[x_1, x_2, \ldots, x_{M-1}, x_M\right]^T, \\ \left[x_2, x_3, \ldots, x_M, x_{M+1}\right]^T, \\ \vdots \\ \left[x_{T-M+1}, x_{T-M+2}, \ldots, x_{T-1}, x_T\right]^T \end{bmatrix} \\ &= \left[d_{x1}, d_{x2}, \ldots, d_{xT-M+1}\right]^T \in R^{(T-M+1)\times M\times m}, \\ D_y &= \begin{bmatrix} \left[y_1, y_2, \ldots, y_{M-1}, y_M\right]^T, \\ \left[y_2, y_3, \ldots, y_M, y_{M+1}\right]^T, \\ \vdots \\ \left[y_{T-M+1}, y_{T-M+2}, \ldots, y_{T-1}, y_T\right]^T \end{bmatrix} \\ &= \left[d_{y1}, d_{y2}, \ldots, d_{yT-M+1}\right]^T \in R^{(T-M+1)\times M\times n} \end{aligned} \tag{4}$$

where T denotes the total moment of data sampling, and m and n denote the number of process variables and quality variables, respectively.

B. Structure and Training Procedure of the MC-CNN

For the design of the MC-CNN, two factors are emphasized. On the one hand, it considers introducing the temporal feature of variables into the model and improving the utilization of the real values. On the other hand, because the traditional CNN is not effective for feature extraction in soft sensor modeling, the MC-CNN is required to not only have the temporal feature between process variables and quality variables, but also have the temporal feature among all the quality variables.

Moreover, the BP process for training is also different from the traditional process. As shown in Fig. 2(a), the normal network can be trained through the traditional BP algorithm. However, for multirate quality variables, blank units will appear during training like the white units shown in

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

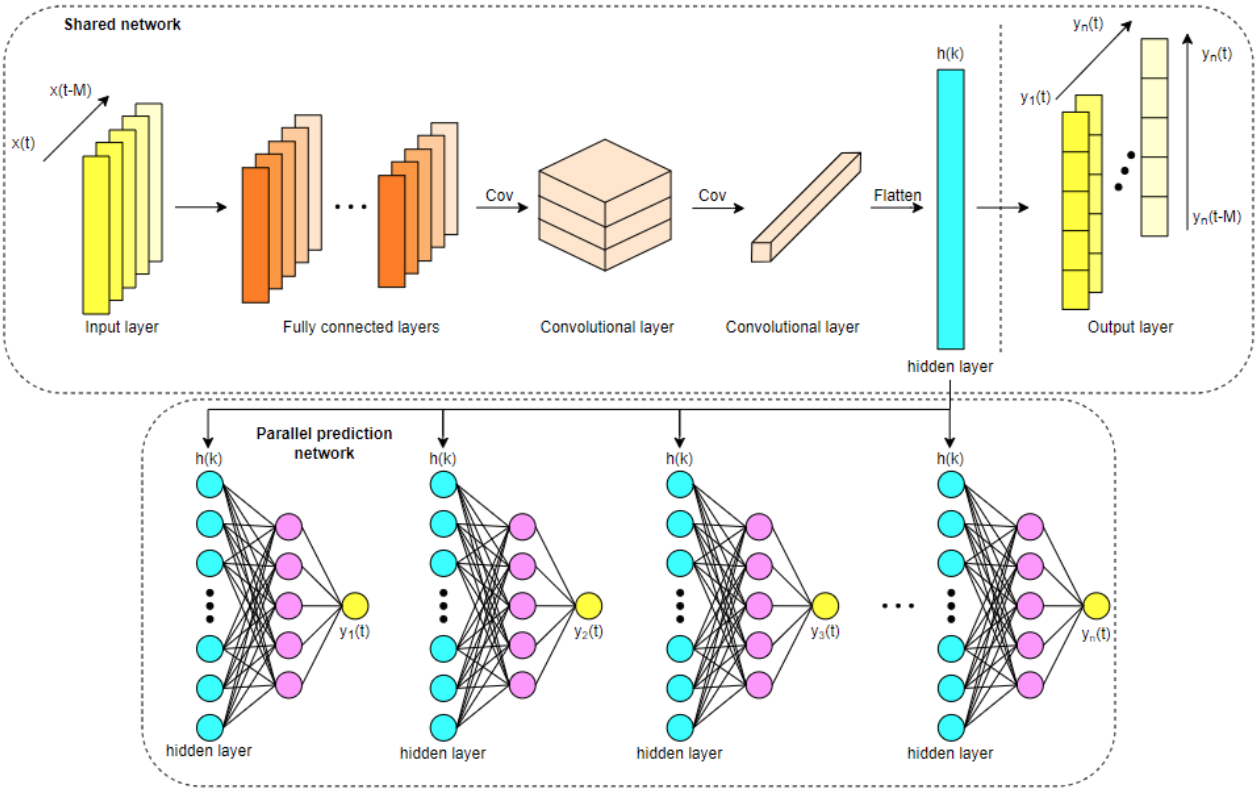

Fig. 3. Structure of the MC-CNN.

Fig. 2(b). At this moment, the network cannot be trained by the traditional BP algorithm because of the wrong error caused by the difference between the blank values and the real values. However, if the blank units and corresponding channels are excluded like what is shown in Fig. 2(c), it can be seen that there almost does not influence the training of the remaining channels. Therefore, a modified BP algorithm is proposed and used. To be more understandable below, the real elements are represented by y_r. In the modified BP algorithm, the blank values will be identified and not participate in the process of backpropagation of the training, so the function is built as

$$g(x-y)=\begin{cases} x-y, & y\in y_r \\ 0, & y\notin y_r \end{cases} \tag{5}$$

the loss function of the network is set as

$$J_s=\frac{1}{2NZ}\sum_{i=1}^{N}\sum_{j=1}^{M}\left\{g\left(\tilde{y}_{(i,j)}-y_{(i,j)}\right)\right\}^2 \tag{6}$$

where $\tilde{y}_{(i,j)}$ denotes the predicted output of the network, Z denotes the number of real values in label, M denotes the number of multi-temporal channels.

It is worth noting that the method proposed in the article is similar to dropout and the difference between dropout and the method proposed can be reflected in two aspects. First, dropout is a kind of regularization to alleviate the overfitting problem, while the method proposed in the article is used to solve the prediction problem of multisampling rates quality variables. Second, dropout is a random process and the dropped units are randomly determined, while the proposed method specifically drops blank units in the last layer. Dropout is different from the method proposed in the article in both the function and the implementation.

The structure of the MC-CNN is divided into two parts: shared network and parallel prediction network. As shown in Fig. 3, in the shared network, the quality variables of M time length are used as labels, and the temporal feature extracted from data passing through multiple fully connected layers and convolutional layers will be flattened to one dimension. It is worth noting that compared with the traditional CNN, the pooling layer is not used in the MC-CNN considering that it may filter out some secondary features, which is against the soft sensor construction. The parameters of convolutional layers like the number of layers, convolution kernel size, and the step size of the receptive field's movement can be determined according to the number of channels M of the network and the dimension of the data. A parallel prediction network contains multiple quality variable prediction networks, and each subnetwork corresponds to a quality variable. The feature extracted from the shared network serves as the common input of these subnetworks to train each network with real-time quality variables as labels, thus completing the construction of soft sensors for all quality variables. Networks are all activated by the SELU function. Compared with other MIMO models, the structure of the proposed network greatly utilizes the limited real values of the quality variables and improves the ability of feature extraction. Meanwhile, the simple network structure and the small number of parameters also

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

ensure that the time required for the network training is not too long.

The training procedure is given as follows.

1) *Data Preparation*: In the network training, the required data form Y_{nb}, D_x, and D_y are introduced in (3) and (4). Among them, all three datasets have elements that will not participate in the training process because of the blank value processing.

2) *Pretraining the Shared Network*: Use dataset like D_x and D_y in form as the input and label, the modified BP algorithm is used for training, and the loss function of the network is set as (6). During training, after a set of input like $[x_1, x_2, x_3, x_4, x_5]^T$ and label like $[y_1, y_2, y_{r3}, y_4, y_5]^T$ are trained, the time window will move a single time step like $[x_2, x_3, x_4, x_5, x_6]^T$ as input and $[y_2, y_{r3}, y_4, y_5, y_6]^T$ as label, while the real values like y_{r3} in the label move to another node and is utilized again. In detail, because there are five multitemporal channels set up in this article, so one real value can be placed at five different units in the output layer of the network. These five situations correspond to five process variable inputs which means that every real value can be used five times. This shows that the MC-CNN can find different temporal relationships between quality variables and process variables and also shows that the model can better extract the feature $h(k)$ represented by the hidden layer.

3) *Fine-Tuning Every Quality Variable Prediction Network*: For each quality variable prediction network, the feature $h(k)$ obtained in the shared network is used as input, and the dataset corresponding to each quality variable is used as the label. Similarly, the loss function for all networks is set as

$$J_p = \frac{1}{2Z}\sum_{i=1}^{N}\left\{g\left(\tilde{y}_{(i)} - y_{(i)}\right)\right\}^2 \tag{7}$$

where $y_{(i)}$ denotes the label and Z denotes the number of real values in label.

C. MC-CNN for Quality Prediction

The MC-CNN can be simply used to build soft sensors. The flowchart of the MC-CNN is shown in Fig. 4. The main steps are listed as follows.

1) Collect process variables X and quality variables Y according to industrial requirements and divide the acquired data into a training set $\{X_{tr}, Y_{tr}\}$ and testing set $\{X_{te}, Y_{te}\}$.

2) Standardize data to get training set $\{X_{n-tr}, Y_{n-tr}\}$ and testing set $\{X_{n-te}, Y_{n-te}\}$.

3) Process the blank values of the data making them not participate in the training and can be identified by the modified BP algorithm. In this article, a special value greatly far away from the standardized real values is filled into the blank values. If the special value is identified during training, it will be excluded from BP. The training set and the testing set completing the processing can be represented as $\{X_{nb-tr}, Y_{nb-tr}\}$ and $\{X_{nb-te}, Y_{nb-te}\}$, respectively.

4) Enhance the dimension of data. For this reason, the quality variable needs to be predicted online, so it does not need to process Y_{nb-te}. The training set and the testing set completing the processing can be represented as $\{D_{x-tr}, D_{y-tr}\}$ and $\{D_{x-te}, Y_{nb-te}\}$, respectively.

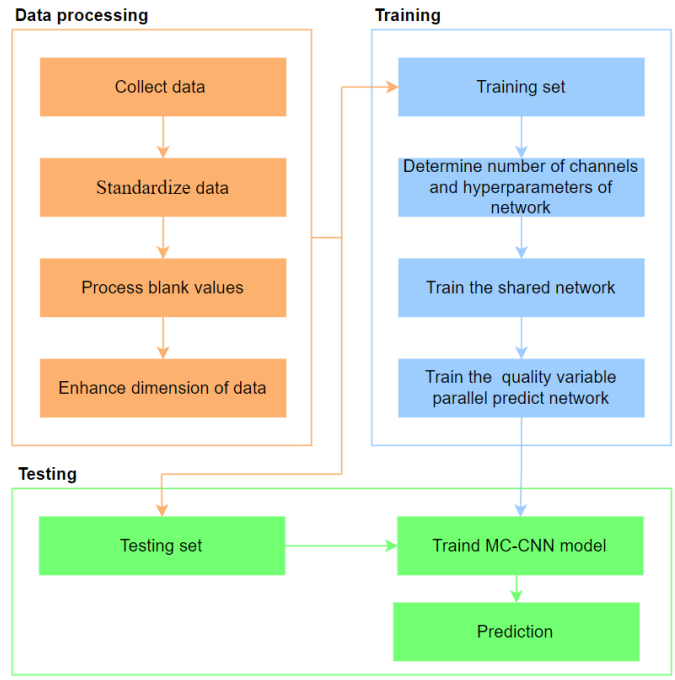

Fig. 4. Flowchart of the MC-CNN for quality prediction.

5) Determine the number of multitemporal channels M and the hyperparameters of the network such as the number of layers, learning rate, epoch, and so on.

6) Use the set $\{D_{x-tr}, D_{y-tr}\}$ as the input and label of the network to train the shared network and extract the feature $h(k)$.

7) Use $h(k)$ and Y_{nb-tr} as the input and label of the network to train a parallel prediction network and get the soft-sensor model for quality variables.

8) The testing set $\{D_{x-te}, Y_{nb-te}\}$ is sent into the trained MC-CNN model for quality prediction.

9) Evaluate the performance of the model by comparing the difference between Y_{nb-te} and the predicted values $Y_{predict}$.

It should be noted that in the actual industrial processes, the input composed of the process variables of the previous $M-1$ moments, and process variables of the latest moment are sent into the shared network and the parallel prediction network to complete the online prediction.

IV. Case Studies

In this section, the proposed MC-CNN method is tested under the TE process and the energy storage data obtained in the Zhenjiang Energy Storage Power Station in China to verify its feasibility and effectiveness. The model performance is evaluated by calculating the root mean square error (RMSE). RMSE reflects the deviation of the predicted output from the real output of the testing set. The smaller the value is, the more accurate the predicted value is. Thus, RMSE can reflect the accuracy of the prediction model. RMSE is calculated as follows:

$$\mathrm{rms}E_i = \sqrt{\frac{1}{N}\sum\nolimits_{j=1}^{N}\left(y_{i(j)} - \tilde{y}_{i(j)}\right)^2} \tag{8}$$

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

TABLE I
HYPERPARAMETERS OF THE MC-CNN MODEL FOR THE TE PROCESS

Training procedures	Hyperparameters		
	Learning rate	Epoch	Batch size
Pre-training	0.01	8	32
Fine-tuning	0.01	6	32

where N denotes the number of samples and $y_{i(j)}$ and $\tilde{y}_{i(j)}$ denote real and predictive values of the ith quality variable, respectively.

A. Description of the TE Process

Tennessee Eastman (TE) process is a reliable simulation program of a chemical plant, which is widely used as a benchmark. The TE process contains 41 process variables and 11 manipulation variables. In addition, the TE process has 21 faults with different fault types. The datasets used in this section are generated using the second control structure listed in [36]. Among them, the dataset includes data from normal working conditions and from fault 1 to fault 11. The training set and the testing set for each working condition contain 480 training samples and 960 testing samples, respectively. The sampling time is 3 min and the fault occurs for every fault from 160. In the experiment, component G and component H are selected as quality variable 1 and quality variable 2, respectively, and 33 variables are selected as process variables. Some of the values in the dataset are artificially removed to meet the data form, and 12 groups of training sets are combined into a training set to verify the effect of the model.

B. Experiments and Results of the TE Process

To establish the MC-CNN model, the first step is to determine the structure and hyperparameters of the network. According to the data dimension and the number of multi-temporal channels, the structure of the shared network for predicting quality variables from the TE process is composed of one fully connected layer, two convolutional layers, and two fully connected layers. The dimensions of layers are [[5,33]-[5,40]-[3,38]-[36]-[25]-[5,2]]. In [m_1,m_2] above, m_1 represents the number of channels and m_2 represents the number of units in each layer. The structure of each quality variable predict network is two fully connected layers and the dimensions of the layers are [25-5-1]. Furthermore, the hyperparameters of the network are obtained from testing. Table I gives the hyperparameters of the MC-CNN model in the TE process.

To verify the effect of the MC-CNN, 1D-CNN, MLP, SAE models, and MR-SAE mentioned above are constructed to predict each two quality variables for two industrial process data and choose appropriate network structures and hyperparameters by testing. Meanwhile, to compare the traditional MIMO temporal model, MIMO-CNN is proposed. MIMO-CNN uses the combination of CNN layers and fully connected layers to extract the temporal feature of process variables and then directly predicts two quality variables at the same time. MIMO-CNN only uses a single channel in the last layer, which means that only labels at the latest moment are used for training. The dimensions of layers in MIMO-CNN are

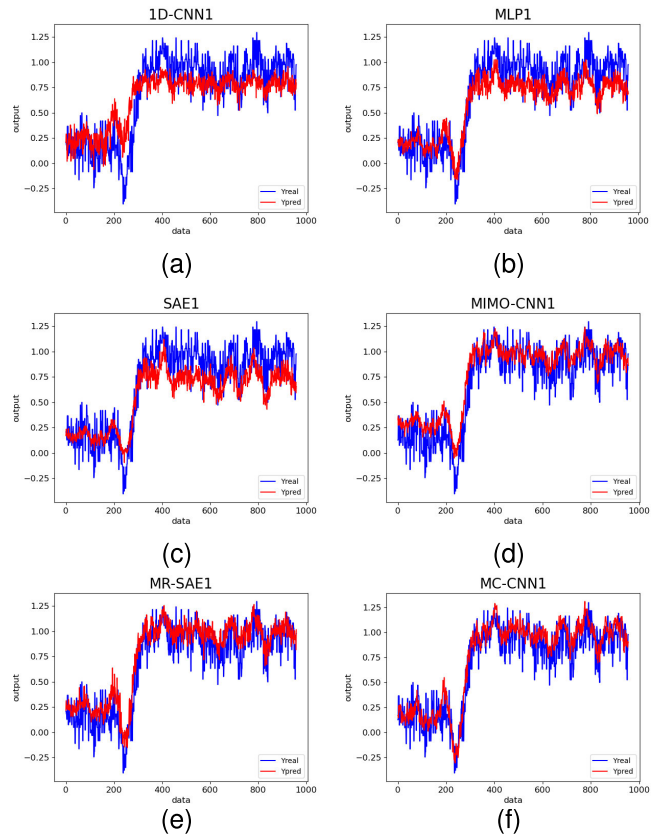

Fig. 5. Predictions on the testing set in fault 2 of component G with six models. (a) 1D-CNN. (b) MLP. (c) SAE. (d) MIMO-CNN. (e) MR-SAE. (f) MC-CNN.

[[5,33]-[5,40]-[3,38]-[36]-[25]-[10]-[2]] for predicting quality variables from the TE process.

Table II shows the testing performance of different models in four working conditions of the TE process. It should be noted that there are four working conditions in the TE process, but only one model of MIMO-CNN, MR-SAE, and MC-CNN is trained to predict quality variable 1 and quality variable 2 at the same time in four working conditions. Meanwhile, two models of the CNN, MLP, and SAE are trained to predict quality variables, respectively. The possible reason why 1D-CNN performs worst in the testing set is that the network may ignore some secondary features during convolution and pooling operations. For this reason, MLP and SAE are better than 1D-CNN. Moreover, MLP has better predictive performance than SAE in most working conditions. MIMO-CNN has the best result on both the quality variables of the TE process except MC-CNN. The reason may be that not only the model is trained on two quality variables simultaneously to extract complementary features, but also the temporal relationship is more important than the mutual relationship among process variables. In contrast, after introducing multiple temporal channels, the MC-CNN has a smaller RMSE in the testing set than other models. It should be added that although Table II only lists the predictive performance of models in four working conditions of the TE process, MC-CNN can guarantee similar or better performance in all other working conditions as well.

Figs. 5 and 6 show the predictive results of the six models for the quality variable 1 (component G) and quality variable 2

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

TABLE II
TESTING PERFORMANCE OF DIFFERENT MODELS IN FOUR WORKING CONDITIONS OF THE TE PROCESS

	TE process							
	Normal		Fault 2		Fault 5		Fault 10	
	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$
1D-CNN	0.1821	0.2792	0.2347	0.3316	0.2060	0.2825	0.1872	0.2905
MLP	0.1715	0.2733	0.2105	0.3056	0.1876	0.2696	0.1767	0.2844
SAE	0.1725	0.2744	0.2147	0.3107	0.1915	0.2720	0.1778	0.2853
MIMO-CNN	0.1605	0.2671	0.1816	0.2791	0.1664	0.2590	0.1640	0.2786
MR-SAE	0.1656	0.2682	0.1856	0.2793	0.1755	0.2588	0.1697	0.2788
MC-CNN	**0.1551**	**0.2638**	**0.1632**	**0.2718**	**0.1596**	**0.2561**	**0.1570**	**0.2751**

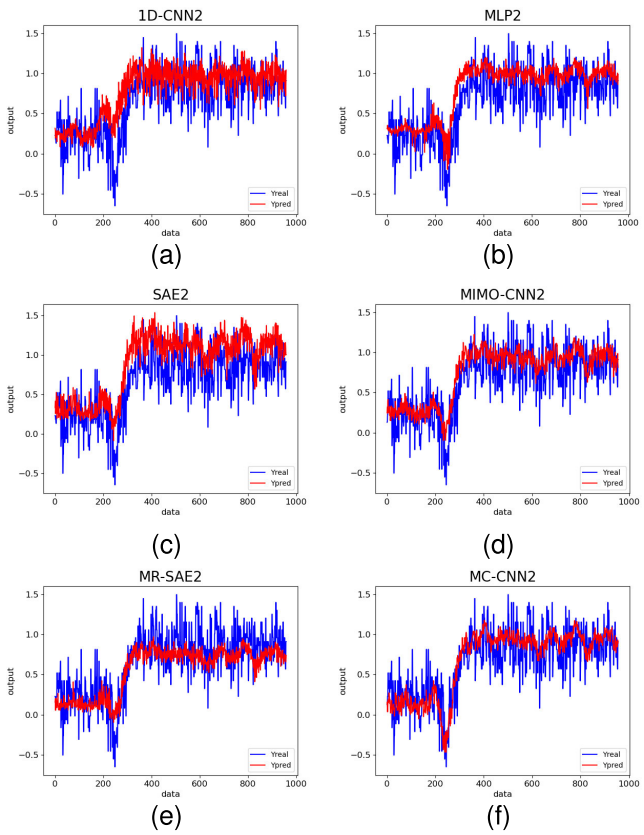

Fig. 6. Predictions on the testing set in fault 2 of component H with six models. (a) 1D-CNN. (b) MLP. (c) SAE. (d) MIMO-CNN. (e) MR-SAE (f) MC-CNN.

(component H) in fault 2 of the TE process. It shows that the proposed MC-CNN model can track the real output curve better compared with other models. In quality variable 1, the MC-CNN can accurately track the curve after the fault occurs. Moreover, in quality variable 2, the MC-CNN not only has the best ability to track the lowest point of the curve, but also can track the curve after the fault occurs, which is obviously superior to other models.

The box plot can intuitively reflect the errors' distribution including the median, the 25th and 75th quartiles, and minimum. Fig. 7 shows the box plots of the absolute values of prediction errors for 1D-CNN, MLP, SAE, MIMO-CNN, MR-SAE, and MC-CNN in the process. It can be seen that the median quartile as well as upper quartile of the MC-CNN

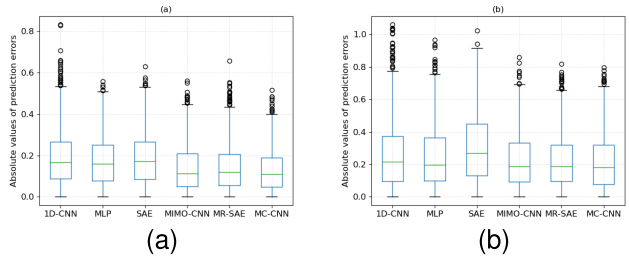

Fig. 7. Box plots of the absolute values of prediction errors on the testing set in fault 2 of components G and H with six methods. (a) Component G. (b) Component H.

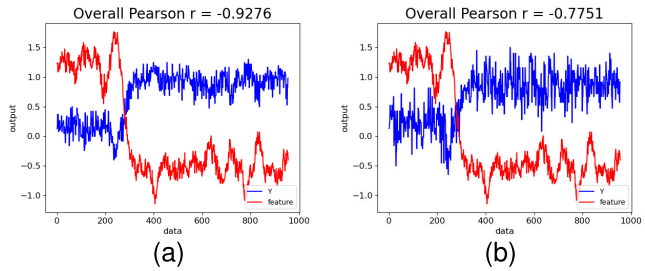

Fig. 8. Time series of features processed by PCA and quality variables in the TE process. (a) Component G. (b) Component H.

model are smaller than other models on the two quality variables, which means that the box plots of MC-CNN have the narrowest error ranges. Hence, the MC-CNN shows better prediction performance than other models.

Fig. 8 shows the time series of features extracted from the shared network, which is processed by PCA to reduce the dimension of the feature and quality variables in the TE process. It shows that there is a strong relationship between feature and quality variables as well a high Pearson correlation shown in the figure also proves that. That also means that the MC-CNN actually can learn the temporal correlations between quality variables and process variables.

C. Description of Data Collected From the Energy Storage Power Station in Zhenjiang of China

By analyzing the energy storage data collected from the Zhenjiang Energy Storage Power Station in China, total cluster voltage and stack voltage are two important variables, which are often difficult to obtain online and have a long time delay. Therefore, they are selected as the quality variable 1 and quality variable 2 of the energy storage data to be

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

TABLE III

HYPERPARAMETERS OF THE MC-CNN MODEL FOR ENERGY STORAGE DATA

Training procedures	Hyperparameters		
	Learning rate	Epoch	Batch size
Pre-training	0.01	8	32
Fine-tuning	0.01	6	32

TABLE IV

TESTING PERFORMANCE OF DIFFERENT MODELS FOR ENERGY STORAGE DATA

	Energy storage data	
	$RMSE_1$	$RMSE_2$
1D-CNN	0.7703	0.5362
MLP	0.7630	0.4193
SAE	0.7537	0.4175
MIMO-CNN	0.7234	0.4682
MR-SAE	0.7186	0.4060
MC-CNN	**0.7165**	**0.3553**

predicted respectively. The historical dataset of energy storage power stations is divided into two categories: battery cell data including voltage, temperature, and the SOC of 228 battery cells, and battery noncell data. More details can be found in [37]. Due to the lack of mechanism analysis of the industrial processes, so 640 of them are selected as process variables for training. In this article, the training set and testing set of the data contain 5760 training samples and 1440 testing samples, respectively.

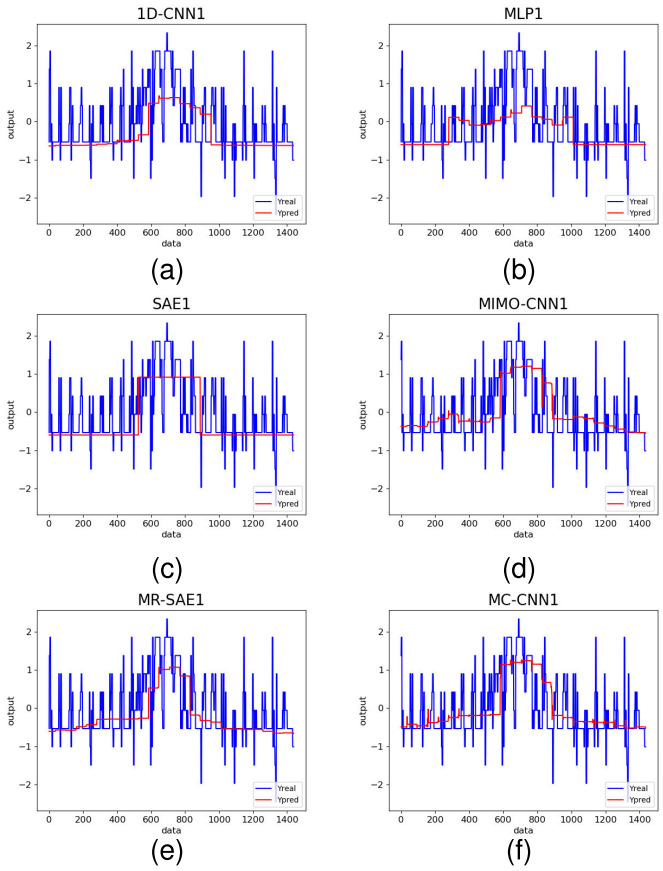

Fig. 9. Predictions on the testing set of total cluster voltage with six models. (a) 1D-CNN. (b) MLP. (c) SAE. (d) MIMO-CNN. (e) MR-SAE. (f) MC-CNN.

D. Experiments and Results of the Energy Storage Data

To establish the MC-CNN model, the first step is to determine the structure and hyperparameters of the network. According to the data dimension and the number of multitemporal channels, the structure of a shared network for predicting quality variables from energy storage data is composed of four fully connected layers, two convolutional layers, and two fully connected layers. The dimensions of layers are [[5,640]-[5,320]-[5,160]-[5,80]-[5,40]-[3,38]-[36]-[25]-[5,2]]. The structure of each quality variable prediction network is two fully connected layers and the dimensions of the layers are [25-5-1]. Table III gives the hyperparameters of the MC-CNN model in energy storage data.

To verify the effect of MC-CNN, 1D-CNN, MLP, and SAE, the proposed MIMO-CNN above and MR-SAE are constructed to predict two quality variables and choose appropriate network structures and hyperparameters. The dimensions of layers in the MIMO-CNN are [[5,640]-[5,320]-[5,160]-[5,80]-[5,40]-[3,38]-[36]-[25]-[5,2]] for predicting quality variables for energy storage data.

Table IV shows the testing performance of different models for energy storage data. 1D-CNN still performs the worst in the testing set. However, different from the result in the TE process, SAE has a better performance than MLP, which may be caused by the higher redundancy of energy storage data. Notably, the MIMO-CNN has a better result on the quality variable 1 of energy storage data. However, for quality variable 2, the prediction accuracy of the model is not ideal. One reason may be that although the model is trained on two quality variables simultaneously to extract complementary features, the model cannot effectively extract features from a large number of process variables due to insufficient knowledge of the energy storage data mechanism. Meanwhile, unlike the result in the TE process, MR-SAE performs better than MIMO-CNN in energy-saving data. The possible reason may also be that a large number of process variables make it difficult to extract the temporal relationship among process variables. In contrast, after introducing multiple-temporal channels, the MC-CNN still has a smaller RMSE in the testing set than other models, which can verify the importance of the temporal correlations between quality variables and process variables to a certain extent.

Figs. 9 and 10 show the prediction results of the six models for quality variable 1 (total cluster voltage) and quality variable 2 (stack voltage) of energy storage data. It can be seen that the proposed MC-CNN model can track the real output curve best compared with other models. The MC-CNN can simultaneously be trained on two quality variables so that the complementary can be extracted. On the other hand, by introducing multitemporal channels, the MC-CNN can extract temporal features and improve the utilization of real values in data as well, allowing the model to explore the different temporal correlations between quality variables and process variables.

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

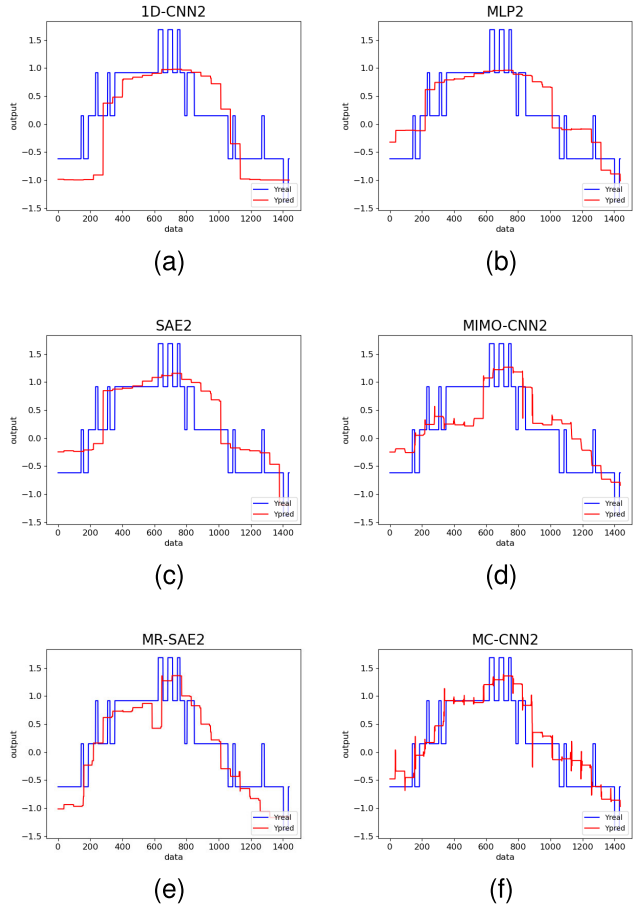

Fig. 10. Predictions on the testing set of stack voltage with six models. (a) 1D-CNN. (b) MLP. (c) SAE. (d) MIMO-CNN. (e) MR-SAE. (f) MC-CNN.

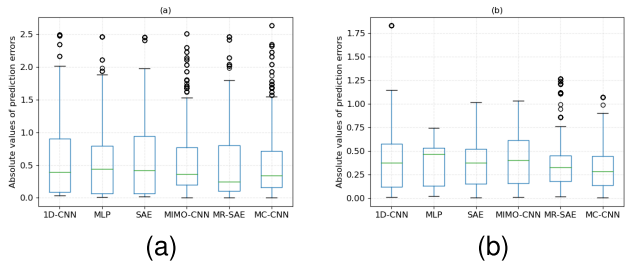

Fig. 11. Box plots of the absolute values of prediction errors on total cluster voltage and stack voltage with six methods. (a) Total cluster voltage. (b) Stack voltage.

Fig. 11 shows the box plots of the absolute values of prediction errors for 1D-CNN, MLP, SAE, MIMO-CNN, MR-SAR, and MC-CNN in energy storage data. It can also be seen that the median quartile as well as upper quartile of the MC-CNN model are smaller than other models on the two quality variables, which means that the box plots of the MC-CNN have the narrowest error ranges. Hence, the MC-CNN can gain better prediction performance than other models.

Fig. 12 shows the time series of features processed by PCA, which is extracted from the shared network and quality variables in energy storage data. It shows that there is a certain

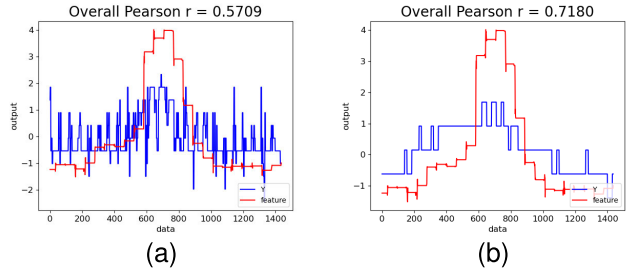

Fig. 12. Time series of features processed by PCA and quality variables in energy storage data. (a) Total cluster voltage. (b) Stack voltage.

relationship between feature and quality variables. The reason why the Pearson correlation is not strong enough maybe that too many process variables influence the feature to learn the temporal correlations between quality variables and process variables. Moreover, the MC-CNN still performs the best in Pearson correlation and time series compared with other models.

The experiments and results above not only verify the ability of the MC-CNN to extract temporal features, but also show that the MC-CNN is the best model to predict multiple quality variables with different sampling rates compared with other models.

E. Ablation Study

Ablation studies can verify the effect of the module by comparing network performance before and after eliminating the module. In the section, to verify the effect of multitemporal channels, a single-temporal channel CNN (SC-CNN) is proposed. The only difference between the SC-CNN and MC-CNN is that the SC-CNN only uses a single channel in the last layer and is trained by the data at the latest moment. The SC-CNN can also be divided into the shared network and the parallel prediction network. The dimensions of the shared network are [[5,33]-[5,40]-[3,38]-[36]-[25]-[2]] and [[5,640]-[5,320]-[5,160]-[5,80]-[5,40]-[3,38]-[36]-[25]-[2]], respectively. The dimensions of each quality variable are still [25-5-1].

Table V shows the testing performance of the SC-CNN and MC-CNN in four working conditions of the TE process and energy storage data. It can be seen that the MC-CNN is always better than the SC-CNN. The reasonable explanation is that without multitemporal channels, the SC-CNN, just like MIMO-CNN mentioned above, can only learn the mutual temporal correlations among process variables rather than the temporal correlations between quality variables and process variables in the situation of mutisampling variables. The ablation study verifies the effectiveness of multitemporal channels and their ability to learn the temporal correlations between quality variables and process variables, which plays an important role in tracking temporal prediction curves.

F. Sensitiveness Analysis of the Parameter M

After verifying the effect of multitemporal channels, how to determine the number of channels M in different industrial

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

TABLE V

TESTING PERFORMANCE OF ABLATION STUDY IN FOUR WORKING CONDITIONS OF THE TE PROCESS AND ENERGY STORAGE DATA

	TE process								Energy storage data	
	Normal		Fault 2		Fault 5		Fault 10			
	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$
SC-CNN	0.1598	0.2687	0.1716	0.2742	0.1649	0.2606	0.1626	0.2812	0.7320	0.4516
MC-CNN	**0.1551**	**0.2638**	**0.1632**	**0.2718**	**0.1596**	**0.2561**	**0.1570**	**0.2751**	**0.7165**	**0.3553**

TABLE VI

TESTING PERFORMANCE OF MODELS TRAINED THROUGH DATA WITH SMALLER AMOUNT FOR THE TE PROCESS AND ENERGY STORAGE DATA

	TE process								Energy storage data	
	Normal		Fault 2		Fault 5		Fault 10			
	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$
MIMO-CNN	0.1677	0.2709	0.1888	0.2809	0.1761	0.2654	0.1707	0.2841	0.7531	0.5279
MR-SAE	0.1712	0.2776	0.1932	0.2867	0.1835	0.2649	0.1712	0.2844	0.7399	0.4737
MC-CNN	**0.1610**	**0.2668**	**0.1720**	**0.2719**	**0.1676**	**0.2603**	**0.1623**	**0.2778**	**0.7261**	**0.4609**

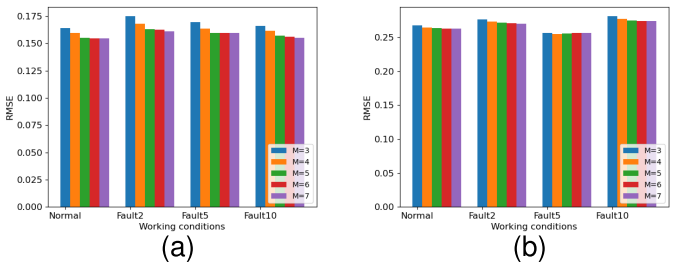

Fig. 13. Prediction performance on components G and H of the MC-CNN with different parameter M in four working conditions of the TE process. (a) Component G. (b) Component H.

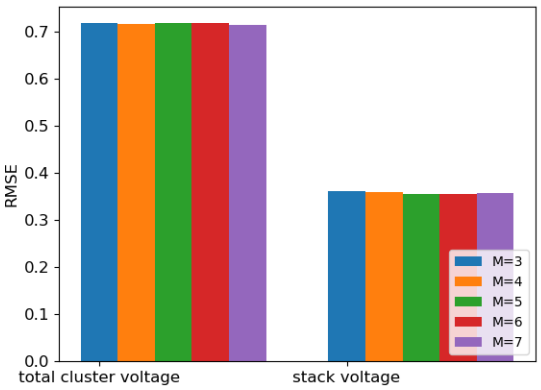

Fig. 14. Prediction performance on total cluster voltage and stack voltage of the MC-CNN with different parameter M of energy storage data.

processes is also worth discussing. To this end, the sensitivity of the parameter M is analyzed in this section. Figs. 13 and 14 show the prediction performance of the MC-CNN with different parameter M in the TE process and energy storage data, respectively. In the TE process, the prediction performance of both components G and H in the four conditions improved significantly when M is changed from 3 to 4. After changing M to 5, the prediction performance of component G is further improved, while the prediction performance of component H remains almost unchanged. Neither the prediction is improved clearly when M is changed to 6 and 7. Similarly, in energy storage data, the prediction performance of stack voltage also is improved obviously before $M = 5$, while the prediction performance of total cluster voltage does not change with M. As can be seen, for these two processes, setting M to 5 can accommodate both the model performance and the training time of the model.

Simply, in the actual industrial process, the selection of M is just similar to choosing an epoch in neural network training. After reaching a certain value of M, the effect of the model will not be significantly improved. The possible explanation is that the model can no longer learn more temporal correlations between quality variables and process variables from new temporal channels. In addition, although a too large value of M will not cause the sharp prediction performance reduction of the model like overfitting, the multiple parameters brought by multiple channels will greatly increase the training time of the model, which is unnecessary. Therefore, it is important to select the appropriate value of M.

G. Experiments and Results With Less Data

To further validate the performance of the MC-CNN with smaller amounts of training data, compare the prediction accuracy of the MIMO-CNN, MR-SAE, and MC-CNN trained by half the amount of data. As can be seen from Table VI, the prediction accuracy of models trained by a smaller amount of training data is declined. The prediction accuracy of the MC-CNN is always better than that of MIMO-CNN and MR-SAE. Moreover, in some situations, the corresponding RMSE values of the MC-CNN trained with half the amount of data are still lower than those of the MIMO-CNN and MR-SAE trained with a normal amount of data. Therefore, it can demonstrate that the MC-CNN has more ability to make full use of low-sampling quality variables and high-sampling process variables through multitemporal channels and also shows that the MC-CNN has stronger adaptability and is more suitable for online prediction of the rapidly changing industrial environment.

H. Generalization Study

By comparing the prediction accuracy of models predicting quality variables from two process industries, MC-CNN shows

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
TESTING PERFORMANCE OF GENERALIZATION STUDY IN FOUR WORKING CONDITIONS OF THE TE PROCESS

	TE process							
	Normal		Fault 2		Fault 5		Fault 10	
	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$	$RMSE_1$	$RMSE_2$
SC-Transformer	0.1854	0.2823	0.2030	0.3028	0.1893	0.3120	0.2082	0.2976
MC-Transformer	**0.1712**	**0.2710**	**0.1830**	**0.2862**	**0.1779**	**0.2766**	**0.1807**	**0.2850**

not only better prediction performance, but also good generalization performance. Meanwhile, to verify the generalization of the proposed module, multi-temporal channels are embedded into the last layer of Transformer [38] as MC-Transformer, which corresponds to SC-Transformer. The structure and hyperparameters of the MC-Transformer and SC-Transformer are obtained from testing. Table VII shows the testing performance of the SC-Transformer and MC-Transformer in four working conditions of the TE process. It can be seen that the MC-Transformer is still better than the SC-Transformer in all four working conditions. The CNN and Transformer perform better after embedding multitemporal channels into models, which proves the generalization of the proposed module.

From the analysis above, we can see that the MC-CNN-based soft sensor not only has a better performance, but a better ability to cope with different amounts of data compared with the traditional 1D-CNN, MLP, and SAE models, but also have good generalization, which makes it more likely to put the MC-CNN used to predict multirate quality variables into practice.

V. CONCLUSION AND OUTLOOKS

In this article, a novel soft sensor based on the MC-CNN is proposed for the prediction of multirate quality variables. The MC-CNN uses multitemporal channels to extract the temporal and complementary feature of the preprocessed data through the shared network of the MC-CNN and then inputs the extracted feature into the subnetwork in a parallel prediction network to train separately for each quality variable. Furthermore, the BP algorithm was modified during the MC-CNN training so that the blank values generated at the unsampled moments are not involved in the BP. At the same time, multitemporal channels are used to increase the utilization of real sampling values. The MC-CNN-based soft sensor is applied to predict components G and H of the TE process as well as total cluster voltage and stack voltage of energy storage data. The result shows that the MC-CNN has lower RMSE values than the CNN, MLP, SAE, MIMO-CNN, and MR-SAE, which means that the MC-CNN outperforms other models in accuracy and reliability. The ablation study proves the effectiveness of the multitemporal channels. The sensitivity analysis of the parameter M shows the way to choose the number of channels. The generalization study shows that the MC-CNN also has great generalization ability. When training models with data with a smaller amount, the prediction result of the MC-CNN still maintained best, proving that the MC-CNN has stronger adaptability and is more suitable for online prediction of the rapidly changing industrial environment.

In this article, the MC-CNN is proposed to solve the problem caused by the high sampling rate of process variables and low sampling rate of multirate quality variables. However, problems like when multirate quality variables maintain low sampling rates, soft sensor construction for process variables at multiple sampling rates, and the inability of the model to focus on specific quality variables remain to be solved. In addition, the improvement of the proposed module and its more suitable application in other DL models and the effect verification for the model trained by irregular sampling rate data are also worth discussing in the future.

REFERENCES

[1] S. Wang, W. Tian, C. Li, Z. Cui, and B. Liu, "Mechanism-based deep learning for tray efficiency soft-sensing in distillation process," *Rel. Eng. Syst. Saf.*, vol. 231, Mar. 2023, Art. no. 109012.

[2] P. Chang, X. Bao, F. Meng, and R. Lu, "Multi-objective pigeon-inspired optimized feature enhancement soft-sensing model of wastewater treatment process," *Expert Syst. Appl.*, vol. 215, Apr. 2023, Art. no. 119193.

[3] F. Harrou, T. Cheng, Y. Sun, T. Leiknes, and N. Ghaffour, "A data-driven soft sensor to forecast energy consumption in wastewater treatment plants: A case study," *IEEE Sensors J.*, vol. 21, no. 4, pp. 4908–4917, Feb. 2021.

[4] S. Gao, S. Qiu, Z. Ma, R. Tian, and Y. Liu, "SVAE-WGAN-based soft sensor data supplement method for process industry," *IEEE Sensors J.*, vol. 22, no. 1, pp. 601–610, Jan. 2022.

[5] V. Gopakumar, S. Tiwari, and I. Rahman, "A deep learning based data driven soft sensor for bioprocesses," *Biochem. Eng. J.*, vol. 136, pp. 28–39, Aug. 2018.

[6] X. Yuan, C. Ou, Y. Wang, C. Yang, and W. Gui, "A layer-wise data augmentation strategy for deep learning networks and its soft sensor application in an industrial hydrocracking process," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 8, pp. 3296–3305, Aug. 2021.

[7] L. Ren, T. Wang, Y. Laili, and L. Zhang, "A data-driven self-supervised LSTM-DeepFM model for industrial soft sensor," *IEEE Trans. Ind. Informat.*, vol. 18, no. 9, pp. 5859–5869, Sep. 2022.

[8] Y. Qin, C. Zhao, and B. Huang, "A new soft-sensor algorithm with concurrent consideration of slowness and quality interpretation for dynamic chemical process," *Chem. Eng. Sci.*, vol. 199, pp. 28–39, May 2019.

[9] G. Wang, Q.-S. Jia, M. Zhou, J. Bi, J. Qiao, and A. Abusorrah, "Artificial neural networks for water quality soft-sensing in wastewater treatment: A review," *Artif. Intell. Rev.*, vol. 55, no. 1, pp. 565–587, Jan. 2022.

[10] A. Torgashov and S. Skogestad, "The use of first principles model for evaluation of adaptive soft sensor for multicomponent distillation unit," *Chem. Eng. Res. Des.*, vol. 151, pp. 70–78, Nov. 2019.

[11] C. Ou et al., "Quality-driven regularization for deep learning networks and its application to industrial soft sensors," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 18, 2023, doi: 10.1109/TNNLS.2022.3144162.

[12] X. Yang, Y. Zhang, Y. A. W. Shardt, X. Li, J. Cui, and C. Tong, "A KPI-based soft sensor development approach incorporating infrequent, variable time delayed measurements," *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 6, pp. 2523–2531, Nov. 2020.

[13] F. Zhang, N. Li, L. Li, S. Wang, and C. Du, "A local semi-supervised ensemble learning strategy for the data-driven soft sensor of the power prediction in wind power generation," *Fuel*, vol. 333, Feb. 2023, Art. no. 126435.

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.

[14] Y. Fan, B. Tao, Y. Zheng, and S.-S. Jang, "A data-driven soft sensor based on multilayer perceptron neural network with a double LASSO approach," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 7, pp. 3972–3979, Jul. 2020.

[15] S. H. Hasnen, M. Shahid, H. Zabiri, and S. A. A. Taqvi, "Semi-supervised adaptive PLS soft-sensor with PCA-based drift correction method for online valuation of NO_x emission in industrial water-tube boiler," *Process Saf. Environ. Protection*, vol. 172, pp. 787–801, Apr. 2023.

[16] P. Lian, H. Liu, X. Wang, and R. Guo, "Soft sensor based on DBN-IPSO-SVR approach for rotor thermal deformation prediction of rotary air-preheater," *Measurement*, vol. 165, Dec. 2020, Art. no. 108109.

[17] A. Abdin, "Applying self-normalizing neural networks to tackle data-driven soft sensing problems in manufacturing lines," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2021, pp. 5770–5773.

[18] X. Wang, W. Hu, K. Li, L. Song, and L. Song, "Modeling of soft sensor based on DBN-ELM and its application in measurement of nutrient solution composition for soilless culture," in *Proc. IEEE Int. Conf. Saf. Produce Informatization (IICSPI)*, Dec. 2018, pp. 93–97.

[19] R. Xie, N. M. Jan, K. Hao, L. Chen, and B. Huang, "Supervised variational autoencoders for soft sensor modeling with missing data," *IEEE Trans. Ind. Informat.*, vol. 16, no. 4, pp. 2820–2828, Apr. 2020.

[20] S. Chang, X. Chen, and C. Zhao, "Flexible clockwork recurrent neural network for multirate industrial soft sensor," *J. Process Control*, vol. 119, pp. 86–100, Nov. 2022.

[21] W. Yu and C. Zhao, "Robust monitoring and fault isolation of nonlinear industrial processes using denoising autoencoder and elastic net," *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 3, pp. 1083–1091, May 2020.

[22] W. Xiong and W. Sun, "Dynamic soft sensor modeling for time sequence data based on convolution filters," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 6, pp. 2696–2703, Nov. 2021.

[23] N. Bargellesi, A. Beghi, M. Rampazzo, and G. A. Susto, "AutoSS: A deep learning-based soft sensor for handling time-series input data," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 6100–6107, Jul. 2021.

[24] K. Wang, C. Shang, L. Liu, Y. Jiang, D. Huang, and F. Yang, "Dynamic soft sensor development based on convolutional neural networks," *Ind. Eng. Chem. Res.*, vol. 58, no. 26, pp. 11521–11531, May 2019.

[25] X.-F. Yuan, L. Li, Y. Shardt, Y.-L. Wang, and C.-H. Yang, "Deep learning with spatiotemporal attention-based LSTM for industrial soft sensor model development," *IEEE Trans. Ind. Electron.*, vol. 68, no. 5, pp. 4404–4414, May 2020.

[26] Q. Li, Q. Du, W. Ba, and C. Shao, "Multiple-input multiple-output soft sensors based on KPCA and MKLS-SVM for quality prediction in atmospheric distillation column," *Int. J. Innov. Comput., Inf. Control*, vol. 8, no. 12, pp. 8215–8230, 2012.

[27] L. Zhou, Y. Wang, Z. Ge, and Z. Song, "Multirate factor analysis models for fault detection in multirate processes," *IEEE Trans. Ind. Informat.*, vol. 15, no. 7, pp. 4076–4085, Jul. 2019.

[28] L. Zhou, J. Chen, J. Jie, and Z. Song, "Multiple probability principal component analysis for process monitoring with multi-rate measurements," *J. Taiwan Inst. Chem. Eng.*, vol. 96, pp. 18–28, Mar. 2019.

[29] B. Shen, L. Yao, and Z. Ge, "Predictive modeling with multiresolution pyramid VAE and industrial soft sensor applications," *IEEE Trans. Cybern.*, vol. 53, no. 8, pp. 4867–4879, Aug. 2023.

[30] X. Yuan, L. Feng, K. Wang, Y. Wang, and L. Ye, "Deep learning for data modeling of multirate quality variables in industrial processes," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021.

[31] W. Yu, C. Zhao, and B. Huang, "MoniNet with concurrent analytics of temporal and spatial information for fault detection in industrial processes," *IEEE Trans. Cybern.*, vol. 52, no. 8, pp. 8340–8351, Aug. 2022.

[32] X. Yuan, S. Qi, Y. Wang, K. Wang, C. Yang, and L. Ye, "Quality variable prediction for nonlinear dynamic industrial processes based on temporal convolutional networks," *IEEE Sensors J.*, vol. 21, no. 18, pp. 20493–20503, Sep. 2021.

[33] S. Albawi, T. A. Mohammed, and S. Al-Zawi, "Understanding of a convolutional neural network," in *Proc. Int. Conf. Eng. Technol. (ICET)*, Aug. 2017, pp. 1–6.

[34] W. Jin, Z. J. Li, L.S. Wei, and H. Zhen, "The improvements of BP neural network learning algorithm, in *Proc. WCC ICSP 5th Int. Conf. Signal Process. 16th World Comput. Congr.*, Aug. 2000, pp. 1647–1649.

[35] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, no. 6088, pp. 533–536, Oct. 1986.

[36] P. R. Lyman and C. Georgakis, "Plant-wide control of the Tennessee Eastman problem," *Comput. Chem. Eng.*, vol. 19, no. 3, pp. 321–331, Mar. 1995.

[37] D. Guo et al., "Multiple elastic networks with time delays for early fault detection and prognostics," *IEEE Access*, vol. 8, pp. 129387–129396, 2020.

[38] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–9.

Authorized licensed use limited to: University of Electronic Science and Tech of China. Downloaded on March 24,2026 at 15:41:18 UTC from IEEE Xplore. Restrictions apply.